

Programme	B. Sc. STATISTICS				
Course Code	STA 8 EJ 415				
Course Title	ANALYSIS OF CLINICAL TRIALS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Different sampling techniques and design of experiments				
Course Summary	To understand different methods to analyze medical data				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Appraise the basic concepts of clinical trials	E	P	Seminar Presentation /
2.	Plan and develop the design of clinical trials	An	C	Instructor-created exams
3.	Determine the sample size in clinical trials	Ap	C	Group Tutorial Work
4.	Conduct bioassays and assimilate the concepts of meta-analysis in clinical trials	Ap	C	Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Basics of Clinical Trials		12	20
	1	Introduction to clinical trials		
	2	The need and ethics of clinical trials, bias and random error in clinical studies		
	3	Protocols, conduct of clinical trials, over view of Phase I-IV trials		
	4	Data management-data definitions, standard operating Procedure		
	5	Informed consent form, case report forms, database design		
	6	Data collection systems for good clinical practice		
II	Design of Clinical Trials		12	15

	7	Design of clinical trials		
	8	Different phases, Comparative and controlled trials, Random allocation, Randomization, response adaptive methods and restricted randomization		
	9	Methods of Blinding, Parallel group designs, Crossover designs, Symmetric designs, Adaptive designs, Group sequential designs		
	10	Zelen's designs, design of bioequivalence trials		
	11	Outcome measures		
III	Sample Size Determination and Testing		12	20
	12	Sample size determination in one and two sample cases		
	13	Comparative trials, activity studies, testing and other purposes		
	14	Unequal sample sizes and case of anova		
	15	Surrogate endpoints-selection and design of trials with surrogate endpoints		
	16	analysis of surrogate end point data		
	17	Reporting and Analysis		
	18	Interpretation of result, multi-center trials		
IV	Meta-Analysis		12	15
	19	Meta-analysis in clinical trials-concept and goals, fixed and random effect approaches		
	20	Bioassay: Direct and indirect assays		
	21	Quantal and quantitative assays		
	22	Parallel line and slope ratio assays, Design of bioassays		
V	Open –Ended		12	
	Problems regarding Module I to Module IV			

Text Book

1. **Friedman, L. M., Furburg, C. D. Demets, L. (1998).** Fundamentals of Clinical Trials, Springer Verlag.
2. **Jennison and Turnbull, B.W. (1999).** Group Sequential Methods with Applications to Clinical Trials, CRC Press.
3. **Kulinskaya, E, Morgeathaler, S and Staudte R G (2008).** Meta-analysis, Wiley.

Reference

1. **Fleiss, J. L. (1989).** The Design and Analysis of Clinical Experiments, Wiley.
2. **Marubeni, E. and M. G. Valsecchi (1994).** Analyzing Survival Data from Clinical Trials and Observational Studies, Wiley and Sons.
3. **Piantadosi S. (1997).** Clinical Trials: A Methodological Perspective. Wiley.
4. **W Rosenberger, J MLachin (2002).** Randomization in Clinical Trials Theory and Practice, Wiley